Name of the Experiment: STUDY ON TEST OF HYPOTHESIS

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Problem-01:

The managing director of an agricultural farm claims that his farm produces 100 times on an average dairy. A random sample of 15 days given the following data set:

110, 118, 130, 135, 142, 146, 112, 100, 80, 88, 96, 112, 123, 124, 130.

It is known that the no of items produced by the farm follows normal distribution with variance 300.

(i) Test the following of 5% level of significance that the average daily production of items of that farm is

a) 100 times

b) More than 100 times and

c) Less than 100 times

Solution:

(a) Variance known and small size (i.e $n<30$) In these situation use the normal test.

Here,

Null hypothesis $H_0: \mu = 100$

Alternate hypothesis $H_1: \mu \neq 100$

Test statistics.

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

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Here, variance, $\sigma^2 = 300$

$\mu_0 = 100$

$n = 15$

Significance level, $\alpha = 0.05$

The critical values of the test statistic, a two tailed test are

$-z_{\alpha/2} = -1.96$ and $z_{\alpha/2} = 1.96$

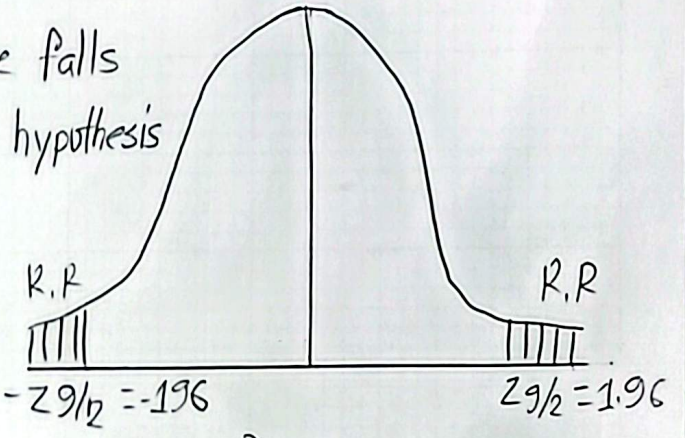
$$\bar{x} = \frac{110+118+130+135+142+146+112+100+80+88+195+112+123+124+130}{15}$$

$$= 116.4$$

$$z: \frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} \sim N(0,1) = \frac{116.4-100}{17.32/3.87}$$

$$= 3.67$$

Comment: As the calculated value falls in rejected area, we consider null hypothesis is rejected.



Or, $z_{0.025} = 1.96$

$-z_{0.023} = -1.96$

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In case of unknown variance,

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t \text{ with } (n-1)$$

$\Sigma x_j^2 = 208482$

$\bar{x} = 116.4$

$$\therefore s^2 = \frac{1}{n-1} \Sigma (x_j - \bar{x})^2$$

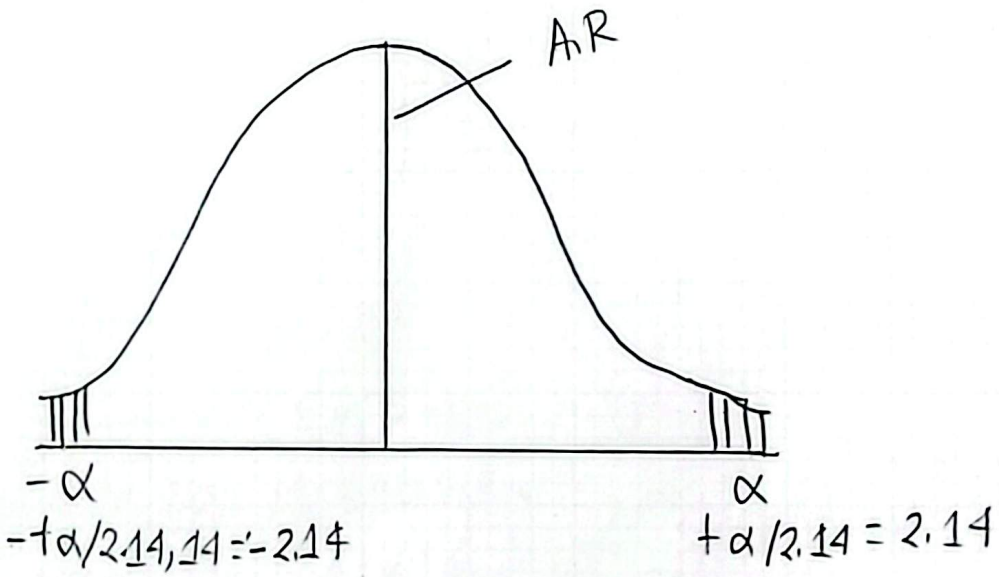
$$= \frac{1}{15-1} \times (208482 - 15 \times 116.4)^2$$

$$= 374.80$$

$$s = 19.36$$

$$\therefore t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$= \frac{116.4 - 100}{19.36/3.273} = 3.28 \sim t \text{ with } 14 \text{ d.f}$$



Comment: As the calculated value falls on critical region, So we consider null hypothesis is significance at level of 5%.

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(b) Solution :

In case of known variance,

Here, Hypothesis, $H_0 : \mu = 100$

$H_1 : \mu > 100$ [One tail test]

Here, $\mu_0 = 100$

$n = 15 \quad \sigma^2 = 300$

Significance level : $\alpha = 0.05$ and $\bar{x} = 116.4$

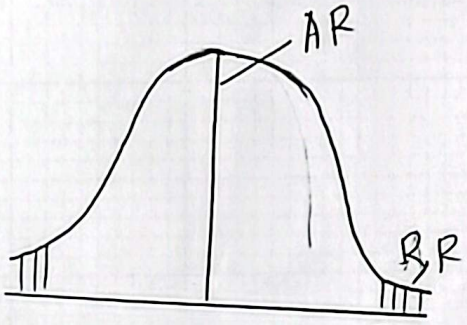
The artificial values of the test statistic a one tailed test,

$-z_\alpha = -1.64$ and $z_\alpha = 1.64$

$\bar{x} = 116.4$

$$Z_z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

$$= \frac{116.4 - 100}{17.32 / 3.87}$$



Comment: As the calculated value falls on critical region we consider null hypothesis is of 5% significance level

In case unknown variance :

$$s^2 = \frac{1}{n-1} \Sigma (x_i - \bar{x})^2$$

$$= \frac{1}{15-1} \times (\Sigma x_i^2 - n\bar{x}^2)$$

$$= 374.80 \quad \therefore s = 19.36$$

$$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}} = \frac{116.4 - 100}{19.36 / 3873}$$

$$= 3.28 \sim t \text{ with } 14 \text{ d.f}$$

$\Sigma x = 20.8482$

$n = 15$

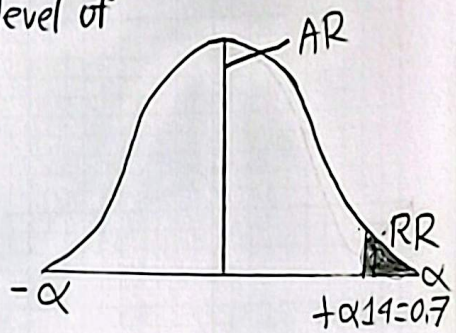
$\bar{x} = 116.4$

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Comment: As the calculated value falls on critical region, we consider null hypothesis is rejected at 5% level of significance.



(c) Solution: In case of known variance, estimated from small sample (i.e. $n < 30$).

Here, Hypothesis, $H_0 : N = 100$
$H_1 : N < 100$

Here, $\mu_0 = 100$
$n = 15$
$\sigma^2 = 300$

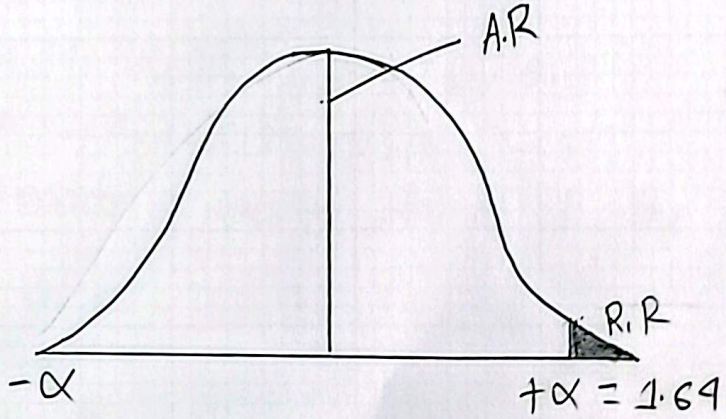
Significance level of $\alpha = 0.05$

The critical values of the test statistic, a one tailed test are $-z_{\alpha, n} = -1.64$ and $z_{\alpha, n} = 1.64$

$\bar{n} = 116.4$

$$z = \frac{116.4 - 100}{17.32/3.87}$$

$$= 3.67$$



Comment: As the calculated value falls on rejected region, we consider null hypothesis is occupied at 5% level of significance.

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In case of unknown variance:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t \text{ with } (n-1) \text{ d.f}$$

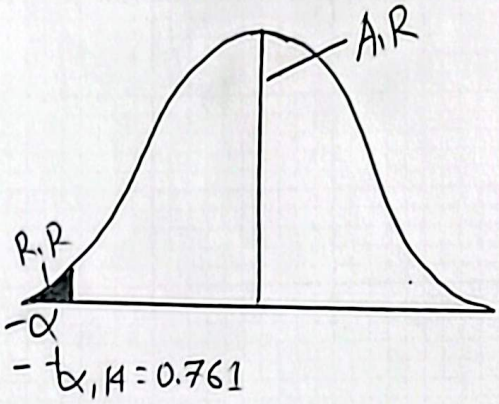
$$s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$$

$$= \frac{1}{15-1} \times 20848 - 15 \times 116.4^2)$$

$$= 374.80, \quad s = 19.36$$

$$t = \frac{116.4 - 100}{19.36/3.873}$$

$$= 3.28 \text{ with } 14 \text{ d.f}$$



Comment: As the calculated value falls on critical region, we consider null hypothesis may be accepted at 5% level of significance.

Problem-02: Two level of fertilizer are tried to compare their effects on the increase in yield of a local rice variety. The following two random samples of observing per ha yield in quantiles.

Level-1 (n_1) : 34, 32, 30, 35, 34, 34, 32, 30, 31, 35, 37 [$n_1 = 11$]

Level-2 (n_2) : 23, 24, 22, 20, 26, 21, 20, 21, 22, 25, 26, 21, 28 [$n_2 = 13$]

Test whether there is any difference is average effects of the two levels of fertilizers.

Solution: $H_0 : \mu_1 = \mu_2$

$H_1 : \mu_1 \neq \mu_2$

Let, $\alpha = 0.05$

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In case of known variance and small sample test statistic.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{1/n_1 + 1/n_2}} \sim t \text{ dist}^n \text{ with } (n_1+n_2-2) \text{ d.f}$$

$$\bar{x}_1 = \frac{\sum x_{1i}}{n_1} = \frac{361}{11} = 38.09$$

$$\bar{x}_2 = \frac{\sum x_{2j}}{n_2} = \frac{299}{13} = 23$$

$$s^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$$

$$s_1^2 = \frac{1}{n_1-1}\sum (x_1-\bar{x})^2$$

$$= \frac{1}{n_1-1} \times \left(\sum x_{1i}^2 - n\bar{x}_1^2\right) = 5.09$$

$$s_2^2 = \frac{1}{n_2-1}\sum (n_{2j}-\bar{x}_2)^2$$

$$= \frac{1}{n_2-1} \times \left(\sum x_{2i}^2 - n\bar{x}_2^2\right) = 6.67$$

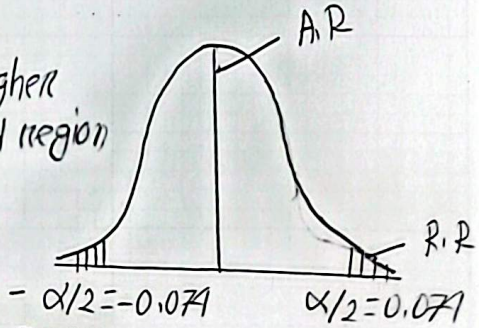
$$s^2 = \frac{(n-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2} = \frac{(11-1)\times 5.09 + (13-1)\times 6.67}{11+13-2}$$

$\therefore s = 2.44$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{1/n_1 + 1/n_2}} = \frac{33.09-23}{2.44 \times \sqrt{1/11+1/13}} = 10.09$$

$= 0.4309$

The critical value of the test statistic a two tailed test are $-t_{\alpha/2,n} = -0.074$ and $t_{\alpha/2,n} = 0.074$



Comment: As the calculated value (10.09) is higher than 2.074. The calculated value fall on rejected region at $\alpha/2 \times 100$ ie. 25% Level of significance with 22 d.f. so, we may concluded that the effect of two fertilizers are different.

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Problem-03: A sample survey of SAU graduate employed in various organization gave the following result:

Types of organization	Utilization of educational qualification presents			
	Fully utilized	Partially utilized	Very little utilized	Total
Education of research	112	19	6	137
Extension	88	27	1	116
Others	27	40	26	93
Total	227	80	33	346

Test whether the type of organization and utilization of education qualification in present job is independent or not.

Soln: H_0 : The attributes are independent

H_1 : The attributes are dependent

Test statistic:

$$\chi^2 = \sum \frac{O_{ij}^2}{E_{ij}} - N \sim \chi^2 \text{ with } (r-1)(c-1) \text{ d.f}$$

Here, $N = 346$

$$E_{ij} = \frac{R_j \times C_i}{N}$$

$$E_{11} = \frac{137 \times 227}{346} = 90, \quad E_{12} = \frac{137 \times 86}{346} \quad E_{13} = \frac{137 \times 33}{346} = 13$$

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$E_{21} = \frac{116 \times 227}{346} = 76$, $E_{22} = \frac{116 \times 86}{346} = 29$, $E_{23} = \frac{116 \times 33}{346} = 11$

$E_{31} = \frac{93 \times 227}{346} = 61$, $E_{32} = \frac{93 \times 86}{346} = 23$, $E_{33} = \frac{93 \times 33}{346} = 9$

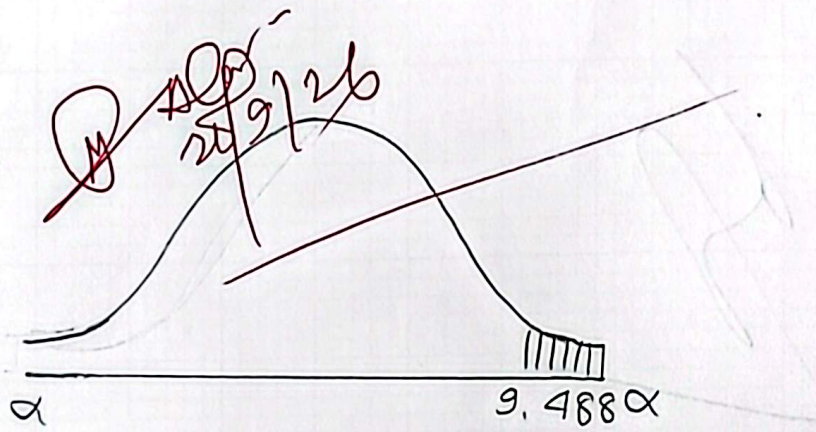
$\therefore \chi^2 = \left(\frac{O_{11}^2}{E_{11}} + \frac{O_{12}^2}{E_{12}} + \frac{O_{13}^2}{E_{13}} + \frac{O_{21}^2}{E_{21}} + \frac{O_{22}^2}{E_{22}} + \frac{O_{23}^2}{E_{23}} + \frac{O_{31}^2}{E_{31}} + \frac{O_{32}^2}{E_{32}} + \frac{O_{33}^2}{E_{33}} \right) - N$

with $\chi^2_{(3-1)(3-1)}$ d.f.

$= \frac{112^2}{90} + \frac{19^2}{34} + \frac{6^2}{13} + \frac{88^2}{76} + \frac{27^2}{29} + \frac{12^2}{11} + \frac{27^2}{61} + \frac{40^2}{23} + \frac{26^2}{9} \right) - 346$

$= 90.52$

The degree of freedom is $(3-1) \times (3-1) = 4$



Comment: In this test, upper tail of the χ^2 dist^n is used as the critical region. The tabulated value for 5% level of significance with 4 d.f. is 9.488. So the calculated value falls on rejected region. Here, the null hypothesis rejected.

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